

AEC Data Extraction from PDF formatted Construction Blueprints

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Abstract

We benchmark six systems on data extraction from real construction drawings under a single uniform condition: one PDF is submitted, one response is returned, and everything the system reports is scored. The benchmark covers two tasks. The first is *object detection* over eight building-element classes — doors by swing type, windows, sinks, toilets, bathtubs, showers and cooktops — comprising 24,771 human-annotated instances. The second is *area and wall segmentation*, comprising 16,112 area and 59,328 wall shapes, scored as pixel masks and, for areas, as instances. The corpus is 312 architectural sheets issued for construction. We evaluate one production document-understanding pipeline (Kamai) against five frontier vision-language models: Gemini 3.1 Pro, Gemini 3.7 Flash, Claude Opus 5, Claude Fable 5 and GPT-5.6 Sol. On the eight object classes every system can emit, scored over byte-identical ground truth, the pipeline reaches F1 0.930 while the strongest model, Gemini 3.7 Flash, reaches 0.062 — a factor of 15. Segmentation is markedly more favourable to the models: they reach 0.612–0.799 area pixel F1 against the pipeline’s 0.983. But area *instance* F1 collapses to 0.078–0.227, meaning they recover area without resolving it into the correct discrete spaces — which is what a quantity takeoff requires. We release fifteen sheets with their ground truth, the frozen class taxonomy with its hash, and per-sheet scores for every system on those sheets.

Keywords: document understanding; object detection; vision-language models; architecture, engineering and construction; benchmark

1 Introduction

A construction drawing is a dense, standardised, information-bearing document. A single architectural sheet routinely carries several hundred discrete symbols — doors by swing type, windows by operation, sanitary and kitchen fixtures — each of which maps to a line item in a quantity takeoff (QTO). Extracting those items is the first step of construction estimating, and it is still performed largely by hand.

Vision-language models are now routinely applied to document understanding, and their reported performance on receipts, forms, charts and slides has improved quickly. Construction drawings are a harder case and a comparatively unmeasured one: instance counts per page are one to two orders of magnitude higher than in typical

document benchmarks, the symbols are small relative to the sheet, and the semantics are governed by drafting convention rather than by natural language.

This paper asks a deliberately narrow question. If a practitioner submits a construction PDF to a system and receives detections back, how good are those detections? We hold the interface fixed — one document in, one response out — and compare a production pipeline built for this document type against five general-purpose frontier models under identical scoring.

Our contributions are: (i) a two-task benchmark of 312 real architectural sheets with 24,771 human-annotated object instances across eight classes plus 16,112 areas and 59,328 wall shapes, an order of magnitude larger in instance count than existing AEC multimodal evaluations; (ii) a scoring protocol that is confidence-free and applies an identical class set and identical ground truth to every system, and whose class taxonomy was frozen and hashed before results were computed; and (iii) a measurement of the resulting gap across both tasks, together with a per-class decomposition.

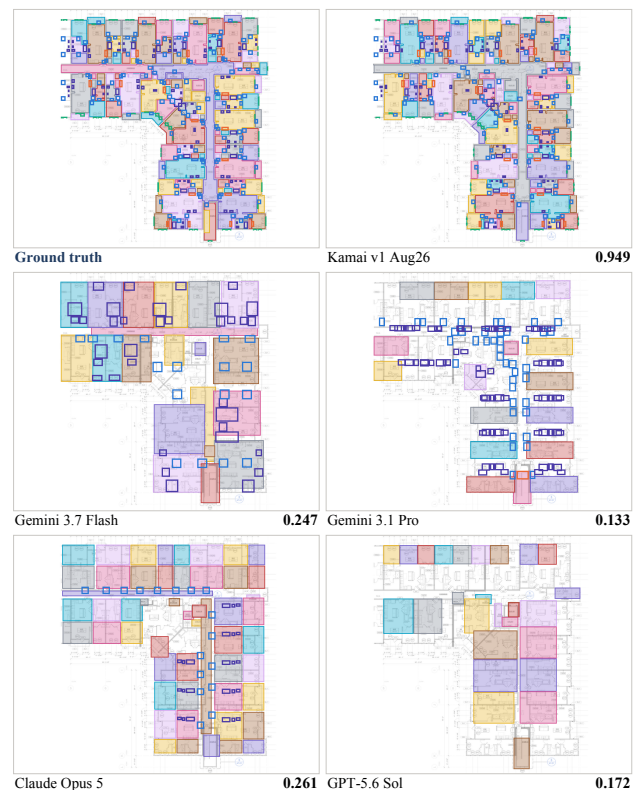


Figure 1: One sheet, six systems (previous page). A released sheet carrying 144 annotated areas and 313 scored objects, shown as ground truth and five systems on the identical crop. Area fills are coloured per enclosed space, object outlines by class. Each score is the mean of that system’s object, wall-pixel, area-pixel and area-instance F1 here. Claude Table 5 is omitted for space; all six are in Appendix A.

Category	Task	Kamai v1 Aug26			Gemini 3.7 Flash			Gemini 3.1 Pro			Opus 5			Fable 5			GPT-5.6 Sol		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Openings	single-swing-door	.971	.959	.965	.152	.053	.079	.048	.023	.031	.097	.041	.057	.135	.027	.046	.037	.012	.018
	double-swing-door	.939	.947	.943	.244	.024	.044	.027	.010	.015	.208	.013	.024	.236	.017	.031	.050	.006	.011
	window	.862	.858	.860	.146	.024	.042	.043	.010	.016	.088	.005	.009	.080	.001	.002	.030	.001	.001
Fixtures	sink	.970	.977	.974	.049	.021	.029	.018	.011	.013	.151	.032	.053	.154	.023	.039	.027	.011	.015
	toilet	.969	.985	.977	.050	.032	.039	.014	.010	.012	.194	.048	.078	.107	.029	.046	.017	.011	.013
	bathtub	.980	.947	.963	.130	.089	.105	.042	.030	.035	.183	.058	.088	.228	.059	.094	.073	.027	.039
	shower	.919	.924	.922	.156	.084	.109	.041	.034	.037	.297	.071	.115	.346	.062	.105	.102	.043	.061
	cooktops	.944	.916	.930	.206	.125	.156	.067	.043	.052	.194	.072	.105	.211	.060	.093	.055	.027	.036
Segmentation	wall-pixel-mask	.943	.927	.935	.222	.329	.265	.226	.123	.159	.236	.311	.268	.230	.284	.254	.204	.043	.071
	area-pixel-mask	.978	.988	.983	.836	.765	.799	.848	.478	.612	.861	.687	.764	.837	.688	.756	.833	.746	.788
	area-instance	.950	.900	.924	.339	.044	.078	.350	.076	.125	.298	.183	.227	.353	.122	.181	.336	.112	.168
All	micro average	.932	.928	.930	.122	.042	.062	.037	.017	.023	.126	.030	.049	.152	.021	.037	.037	.010	.015

Table 1: AEC data extraction, per task. Precision, recall and F1 for every system. Object classes are matched at IoU 0.50; wall and area rows are pixel masks and area-instance is matched at IoU 0.50. The F1 column is shaded for every system; the best F1 in each row is set in blue. Micro average is over the eight object classes. Kamai is the production pipeline; the remaining five are frontier vision-language models.

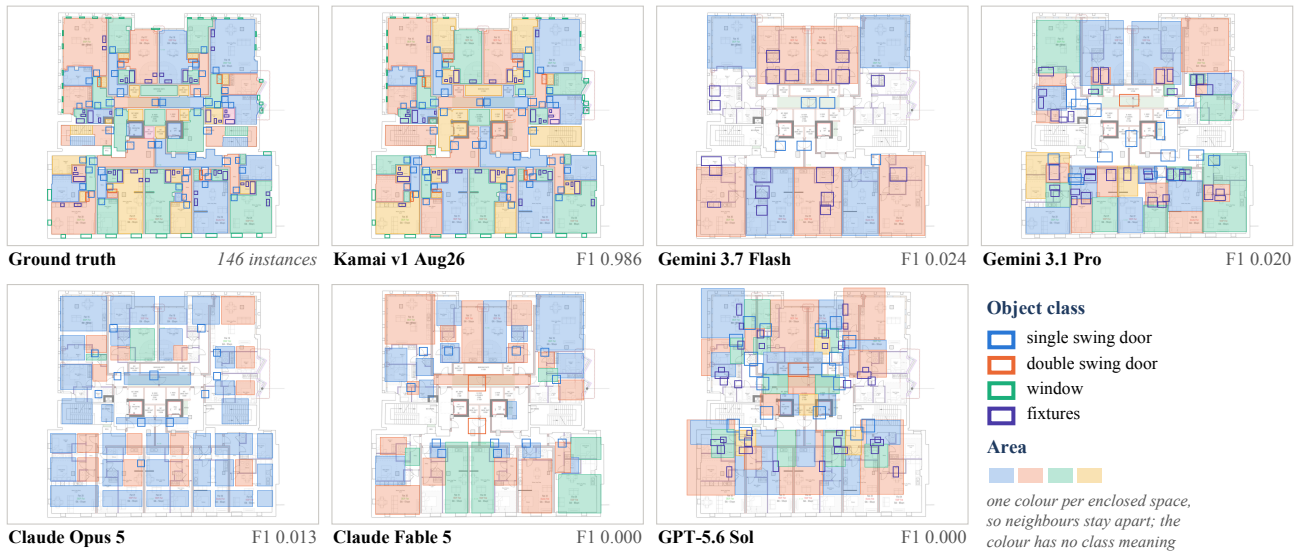


Figure 2: One sheet, every system. Ground truth against all six systems on the same drawing and the same crop — the page’s scored region, which excludes the titleblock the scorer already treats as a void. Object outlines are coloured by class; area fills are coloured per enclosed space so that neighbours stay apart, where an area covers rooms, shafts, balconies, elevator cores and stairs. The pipeline reproduces the annotation almost exactly (F1 0.986 on this page). Claude Fable 5 and GPT-5.6 Sol do return boxes here — 14 and 53 of them — and still score 0.000, so on this sheet the failure is misplacement rather than silence. This drawing is one of the 15 released sheets, shown redacted exactly as published.

2 Related Work

Floor-plan understanding. Raster floor-plan parsing has an established literature, with CubiCasa5K [1] and Raster-to-Vector [2] the most-used corpora. Neither is a construction document set. CubiCasa5K comprises 5,000 scanned plans drawn mainly from Finnish real-estate marketing material, annotated with 83 categories that its benchmark collapses to twelve room and eleven icon classes; Raster-to-Vector annotates 870 plans sampled from a Japanese property-listing portal with eight icon and eleven room types, treats doors and windows as a single undifferentiated opening class, and does not redistribute the images. Both consist of single residential presentation plans without dimensions, notes, schedules or sheet context. FloorPlanCAD [3] originates in production CAD from more than 100 real residential and commercial projects, but is released as 10 m by 10 m vector tiles with text, dimensions and commercially sensitive fields removed, its design phase unstated, and without quantity or area annotation. None of these corpora pairs full sheets issued for construction with takeoff-oriented labels, which is what a quantity takeoff benchmark requires.

Multimodal benchmarks in AEC. AEC-Bench [4] packages complete public-sector bid, permit and construction-document sets into 196 coordination and review tasks. The documents are real issued sets, but the benchmark carries no element-level or quantity annotation, and its task count suits reasoning-style evaluation rather than per-class detection statistics over a long-tailed symbol vocabulary. The two are complementary: AEC-Bench measures navigation and review across a drawing set, this benchmark measures element-level extraction from the sheet.

VLMs as detectors. Frontier models substantially underperform specialist detectors on dense localisation, and the deficit grows with instance count and shrinks with object size. Our results are consistent with that literature; the contribution here is the magnitude on a specific, economically significant document type, and the observation that the deficit is concentrated rather than uniform.

3 Benchmark

3.1 Corpus

The corpus is 312 single-page architectural sheets issued for construction, spanning residential and commercial projects and a range of sheet sizes, largely ISO A-series. Sheets were annotated in CVAT by a professional annotation team. Ground truth comprises 24,771 scored object instances across eight classes, 59,328 wall shapes and 16,112 area shapes.

Two annotation decisions materially affect what can be claimed. First, a machine-generated *Opening* class present in the export is excluded in full: all 9,722 of its shapes carry `source="auto"`, so scoring against it would measure agreement with an algorithm rather than with a person. Second, all window subtypes are collapsed into a single *Window* class on both the ground-truth and prediction

sides, since subtype boundaries are not consistently separable and their distinction is not required by takeoff.

3.2 Task and class set

The task is: given the PDF, return every instance of the target vocabulary with a bounding region and a class label. Eight object classes are scored — Single Swing Door, Double Swing Door, Window, Sink, Toilet, Bathtub, Shower and Cooktops — together with Wall and Area, which are evaluated as segmentation. *Wall* covers walls and railings. *Area* is any enclosed space and covers rooms, shafts, balconies, elevator cores and stairs. Both definitions apply identically to the ground truth and to every prediction. The eight are the elements a quantity takeoff prices: openings and sanitary or kitchen fixtures, each mapping to a schedule line. Every system is scored on all eight, over the same ground-truth instances.

Results are computed over all 312 sheets, carrying 24,771 object instances. Classes present in the annotation but outside the scored vocabulary are excluded from every system’s denominator alike.

4 Evaluation Protocol

Uniform interface. Every system receives the same artefact — the source PDF — and answers with a single response per page. No system receives task-specific decomposition, tiling, retries or scaffolding that the others do not.

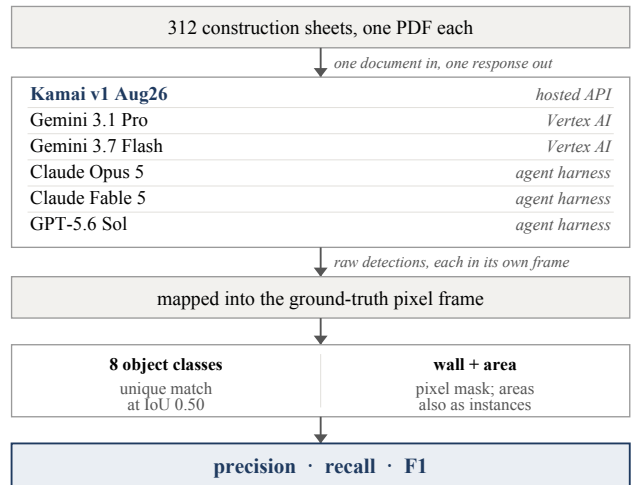


Figure 3: The benchmark protocol. Every system receives the same document and returns one response, which is mapped into the ground-truth pixel frame before scoring. Objects are matched one-to-one at IoU 0.50; walls and areas are scored as pixel masks, areas additionally as instances.

Matching and metric. Objects are matched one-to-one against ground truth at IoU 0.50 and scored as precision, recall and F1. The metric is confidence-free by design: the systems do not emit comparable calibrated scores — some emit none at all and the rest emit uncalibrated self-reported values — and average precision degenerates to precision × recall on score-free output, which would make a ranking metric report something other than ranking. Walls and areas are scored as pixel masks, because ground-truth wall runs are tiled into many

adjacent boxes and instance matching over them is unstable.

Adjacent instances. *Area* is scored as a single class, and any two area instances that touch are merged into one before instance matching — applied identically to the ground truth and to every prediction. This follows the annotation convention: a space is often drawn as several adjoining pieces, and an elevator or shaft is drawn over the area enclosing it, so an unmerged count would record several instances where one space exists. The merge reduces the ground-truth area count from 16,112 to 14,098 (12.5 %). It is not neutral between systems: it raises a system whose regions tile the plan as the annotation does, and lowers one that returns a few coarse regions, because those coalesce further. *Area pixel* F1 is unaffected, since the union is identical.

Identical denominators. Every system is scored on the same eight classes over the same ground-truth instances, so no result depends on a class being excluded for one system and not another. Denominators were verified identical class by class before scoring.

Pre-registration. The class taxonomy and the mapping from each system's output vocabulary to the scored vocabulary were frozen and hashed before any full results were computed; the taxonomy hash (SHA-256 e4bdca9f28224b33) is published with the release. Supplied the ground truth as its own prediction, the harness returns F1 1.000 on every scored class, so a perfect predictor is scored as perfect: this is the check a class-map error would fail.

5 Systems Evaluated

Kamai v1 (Aug26) is the production construction takeoff pipeline behind Kamai (kamai.io). The same in-house geometric foundational models, which read the drawing's vector geometry rather than a rasterised image, serve a self-serve takeoff app (app.kamai.io), a REST API (api.kamai.io) and an MCP server (mcp.kamai.io). Here it is accessed as the hosted API: a PDF is uploaded and a structured takeoff, counts, areas and perimeters per element as JSON, is returned. It is the shipping product, evaluated exactly as a customer would receive it. All results below are for the version of the service in production on 26 August 2026; the pipeline is under active development and later versions will differ.

Gemini 3.1 Pro and **Gemini 3.7 Flash** were accessed through Vertex AI, one request per page, with structured output and the vendor's documented coordinate convention. **Claude Opus 5** and **Claude Fable 5** were accessed through an agent harness, one isolated context per page, with an identical prompt and class vocabulary. **GPT-5.6 Sol** was accessed the same way, through the Codex agent harness, one isolated session per page. The three agent-harness systems were shown the page pre-rendered to the identical 2000-pixel-long-edge frame and were told that frame explicitly, so no system had to infer the resolution it was given.

All five models received the same instruction set: a closed 25-string vocabulary, the annotation conventions

used by the human annotators, and an explicit instruction not to report instances they could not resolve.

6 Results

6.1 Object detection

Table 1 reports every class for every system. All six are scored on the same eight classes, over the same 312 sheets and the same 24,771 ground-truth instances; denominators were verified identical class by class.

Kamai returns micro F1 0.930 against 0.062 for the strongest model, Gemini 3.7 Flash, a factor of 15.0. It exceeds 0.85 on every object class; no model exceeds 0.156 on any.

The deficit is overwhelmingly a recall deficit. Model precision ranges 0.037–0.152 and recall 0.010–0.042; false negatives outnumber true positives by between 23:1 and 101:1. The models are not flooding the page with spurious symbols — they are reporting very little, and are right about a minority of what they do report.

The gap is broadly uniform across the object vocabulary. The models are least bad on cooktops (0.036–0.156) and bathtubs (0.035–0.105), both comparatively large and distinctive symbols, and worst on windows, where 4 of the five fall below 0.02.

6.2 Area and wall segmentation

Areas and walls are the second task and are scored as segmentation rather than detection. Walls are reported as pixel-mask F1 only: ground-truth wall runs are tiled into many adjacent boxes, so instance matching over them is unstable and would measure annotation granularity rather than performance. Areas are reported both ways — as a pixel mask and as instances matched at IoU 0.50 — because a takeoff needs the correct number of discrete spaces, not merely the correct floor area.

System	Wall px	Area px	Area inst	Δ
Kamai v1 Aug26	0.935	0.983	0.924	−0.059
Gemini 3.7 Flash	0.265	0.799	0.078	−0.721
GPT-5.6 Sol	0.071	0.788	0.168	−0.620
Claude Opus 5	0.268	0.764	0.227	−0.537
Claude Fable 5	0.254	0.756	0.181	−0.575
Gemini 3.1 Pro	0.159	0.612	0.125	−0.487

Table 2: Area and wall segmentation. Wall and area pixel F1, and area instance F1 at IoU 0.50. Δ is the drop from area pixel F1 to area instance F1 — the cost of resolving extent into discrete spaces.

Segmentation is where the models are closest to usable. Gemini 3.7 Flash reaches 0.799 area pixel F1 against the pipeline's 0.983 — a ratio of 0.81, compared with 0.07 on object detection. An area is a large, closed, high-contrast region, and finding its extent is a materially easier problem than finding a 30-pixel door symbol among several hundred.

The instance column is the caution. Every model loses between 0.49 and 0.72 F1 moving from pixel to instance, against 0.06 for the pipeline. They recover approximately the right floor area while failing to partition it into the right spaces — merging adjacent spaces, or splitting one. For a visual summary that is a modest error; for a schedule

of areas it is disqualifying, because the row count is wrong before any area is computed.

Walls sit between the two regimes: 0.071–0.268 for the models against 0.935 for the pipeline. Wall runs are long, thin and interrupted by openings, and the models tend to return a few coarse strokes where the ground truth carries many short segments. GPT-5.6 Sol is the outlier at 0.071: it frequently declined to trace walls at all, reporting in its own notes that wall coverage would be partial and omitting the class rather than approximating it.

6.3 A released subset

Fifteen of the 312 sheets are released in full for public benchmarking: the source PDFs, 300 dpi renders with the ground truth drawn on top, and a CVAT export of exactly those 15 sheets. They span the range of sheet density in the corpus rather than clustering at one size. The released sheets are redacted: everything outside the annotated drawing region is removed, along with all annotations, link targets and document metadata. The redaction procedure is documented with the release, which is licensed CC BY-NC 4.0 for the drawings and Apache 2.0 for the evaluation code.

System	P	R	F1	Wall px	Area px	Area inst
Kamai v1 Aug26	0.940	0.919	0.929	0.931	0.987	0.941
Gemini 3.7 Flash	0.074	0.020	0.031	0.238	0.797	0.077
Gemini 3.1 Pro	0.025	0.010	0.015	0.182	0.548	0.116
Claude Opus 5	0.061	0.015	0.024	0.278	0.782	0.227
Claude Fable 5	0.066	0.007	0.013	0.287	0.780	0.186
GPT-5.6 Sol	0.007	0.002	0.003	0.089	0.741	0.096

Table 3: The released 15-sheet subset. Eight object classes over 1632 ground-truth instances on the 15 released sheets, scored exactly as in Table 1 — all window subtypes merged into a single Window class on both sides — so every column is directly comparable.

The ordering is unchanged: 0.929 for the pipeline against 0.031 for the strongest model on this subset, Gemini 3.7 Flash. These sheets are harder for the pipeline than the corpus average — 0.929 here against 0.930 in Table 1 — and windows are where the difference sits: 0.860 on the subset against 0.860 corpus-wide. It is released with its ground truth so the comparison can be checked rather than taken on trust. Fifteen sheets is a small sample and the per-class cells rest on few instances.

7 Cost

Per-page inference cost for the model arms accessed by API is \$0.0225 for Gemini 3.7 Flash and \$0.0572 for Gemini 3.1 Pro at published list rates, with mean end-to-end latency of 31.1 and 29.4 seconds respectively. All 624 scored API calls — one per page per API arm — returned successfully with no refusals and no truncated responses. The pipeline is not a language model and has no token bill, so no dollar figure is derivable and none is reported. The Claude arms ran in a harness that records neither tokens nor latency; no estimate is substituted for either.

8 Limitations

This benchmark was produced by the vendor of one of the systems evaluated. We state the following so a reader does not have to recover them.

- **Input access is not equivalent.** The pipeline is a document-understanding system with access to the PDF's internal structure; the models are given the document and interpret it visually. The comparison measures delivered detections under a common interface, not equivalent perceptual conditions.
- **Two arms lack instrumentation.** The Claude arms ran through an agent harness rather than the API and record neither token counts nor latency, so their cost columns are empty.
- **Single-annotator ground truth.** The annotation task carried no overlap, so inter-annotator agreement is unmeasured and label noise is unquantified.
- **Training-data provenance.** None of these sheets was used to train the pipeline evaluated here, so the result is not a measurement on seen data. No seen/unseen partition is therefore computed or reported. That exclusion rests on the vendor's own records and is not independently auditable from the released material.

9 Conclusion

Under a uniform interface — one PDF submitted, one response scored — a production pipeline returns F1 0.930 on construction-drawing object detection while the strongest of five frontier vision-language models returns 0.062 over identical ground truth. That a purpose-built system outperforms a general-purpose one is unsurprising; the magnitude is. The deficit is driven by recall and concentrated in the small repeated elements that constitute a quantity takeoff — on windows, one model reported 84 instances where the corpus contains 8,042.

Segmentation tells a different and more encouraging story. On enclosed area the models reach 0.612–0.799 pixel F1, against the pipeline's 0.983. The obstacle there is not perception but individuation: area instance F1 falls to 0.078–0.227, so the correct area is recovered without the correct spaces. That is a narrower and more tractable failure than the object-detection gap, and it suggests areas and walls are the part of AEC extraction where general models are closest to practical use.

For practitioners the implication is direct: submitting a construction drawing to a frontier model and scoring what returns does not currently yield a usable takeoff, and the failure is silent — the models return confident, well-formed, largely empty answers rather than declining. For the research community, construction documents remain a document class where general multimodal capability has not transferred, and where the available benchmarks are small relative to the density of the medium. We release fifteen sheets with their ground truth, the frozen taxonomy and per-sheet scores to make the gap tractable.

Appendix A Five released sheets, every system

Each block is one of the released sheets: ground truth first, then all six systems on the identical crop. Object outlines are coloured by class — **single swing door**, **double swing door**, **window**, **fixtures**. Area fills are coloured *per instance* so that one enclosed space is distinguishable from the space beside it; the colour carries no class meaning. Sheets are ordered by ground-truth density.

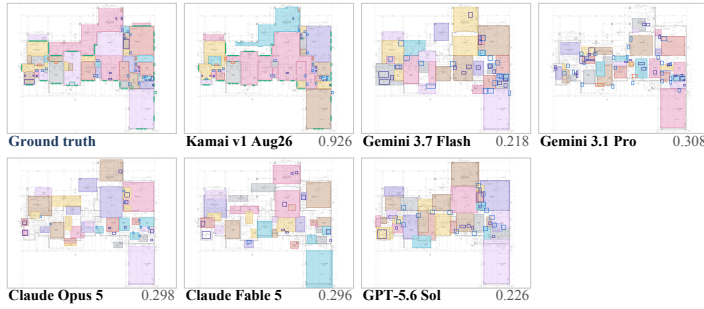


sheet 02

37 objects, 24 areas in ground truth

	obj	wall	area px	area inst	mean
Kamai v1 Aug26	.99	.92	.99	.83	0.930
Gemini 3.7 Flash	.08	.63	.89	.34	0.488
Gemini 3.1 Pro	.00	.00	.89	.70	0.398
Claude Opus 5	.04	.43	.84	.35	0.414
Claude Fable 5	.04	.29	.81	.37	0.378
GPT-5.6 Sol	.00	.34	.89	.57	0.450

mean of the four task scores, this sheet only

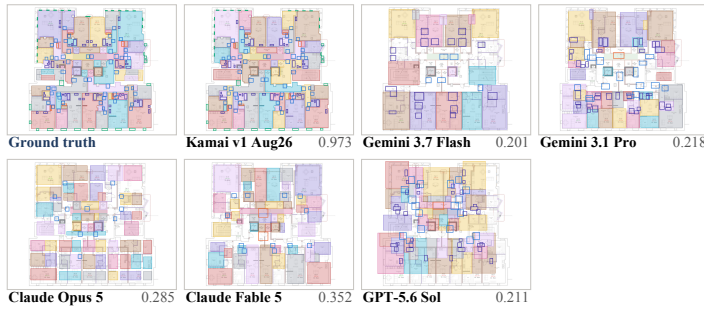


sheet 09

81 objects, 41 areas in ground truth

	obj	wall	area px	area inst	mean
Kamai v1 Aug26	.93	.90	.96	.91	0.926
Gemini 3.7 Flash	.00	.00	.87	.00	0.218
Gemini 3.1 Pro	.00	.31	.74	.19	0.308
Claude Opus 5	.00	.25	.78	.16	0.298
Claude Fable 5	.00	.21	.76	.21	0.296
GPT-5.6 Sol	.02	.00	.88	.00	0.226

mean of the four task scores, this sheet only

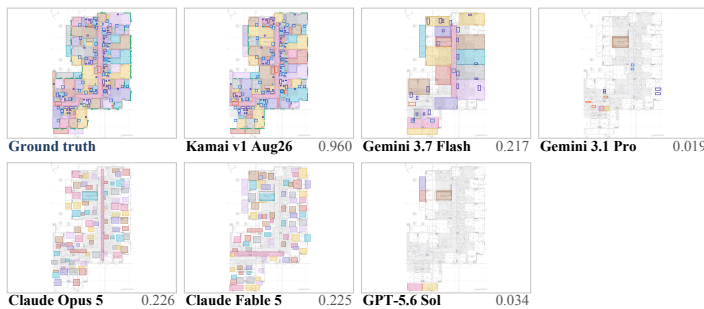


sheet 12

146 objects, 73 areas in ground truth

	obj	wall	area px	area inst	mean
Kamai v1 Aug26	.99	.91	.99	1.00	0.973
Gemini 3.7 Flash	.02	.00	.75	.03	0.201
Gemini 3.1 Pro	.02	.00	.80	.06	0.218
Claude Opus 5	.01	.26	.76	.11	0.285
Claude Fable 5	.00	.34	.79	.28	0.352
GPT-5.6 Sol	.00	.11	.74	.00	0.211

mean of the four task scores, this sheet only

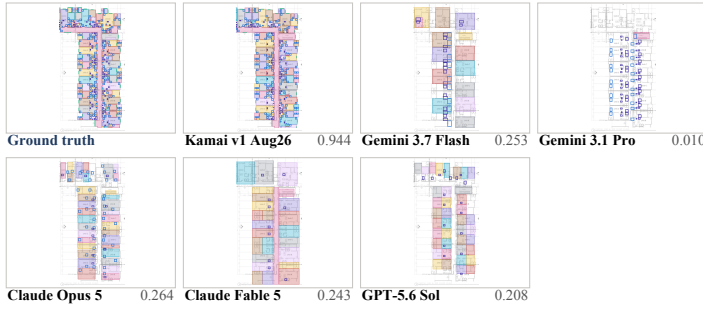


sheet 13

170 objects, 99 areas in ground truth

	obj	wall	area px	area inst	mean
Kamai v1 Aug26	.92	.96	.98	.97	0.960
Gemini 3.7 Flash	.04	.00	.79	.04	0.217
Gemini 3.1 Pro	.00	.00	.05	.02	0.019
Claude Opus 5	.00	.18	.58	.14	0.226
Claude Fable 5	.00	.23	.58	.09	0.225
GPT-5.6 Sol	.00	.00	.10	.04	0.034

mean of the four task scores, this sheet only



sheet 14

308 objects, 208 areas in ground truth

	obj	wall	area px	area inst	mean
Kamai v1 Aug26	.95	.92	.99	.92	0.944
Gemini 3.7 Flash	.00	.27	.73	.01	0.253
Gemini 3.1 Pro	.00	.00	.04	.00	0.010
Claude Opus 5	.00	.18	.73	.14	0.264
Claude Fable 5	.00	.12	.85	.00	0.243
GPT-5.6 Sol	.00	.00	.75	.08	0.208

mean of the four task scores, this sheet only

Figure 4: Five released sheets, every system. Ground truth against all six systems on five of the fifteen released sheets, ordered by ground-truth density; these five carry 37 to 308 annotated objects. Each row gives object, wall pixel, area pixel and area instance F1 for that sheet, and the bar shows their mean, so a system is not judged on objects alone: a model that recovers area as one blob rather than as rooms still scores on the pixel column. Across these five that mean runs 0.926–0.973 and the best model on each sheet runs 0.226–0.488. These five were chosen for density spread and legibility, not sampled at random.

References

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